



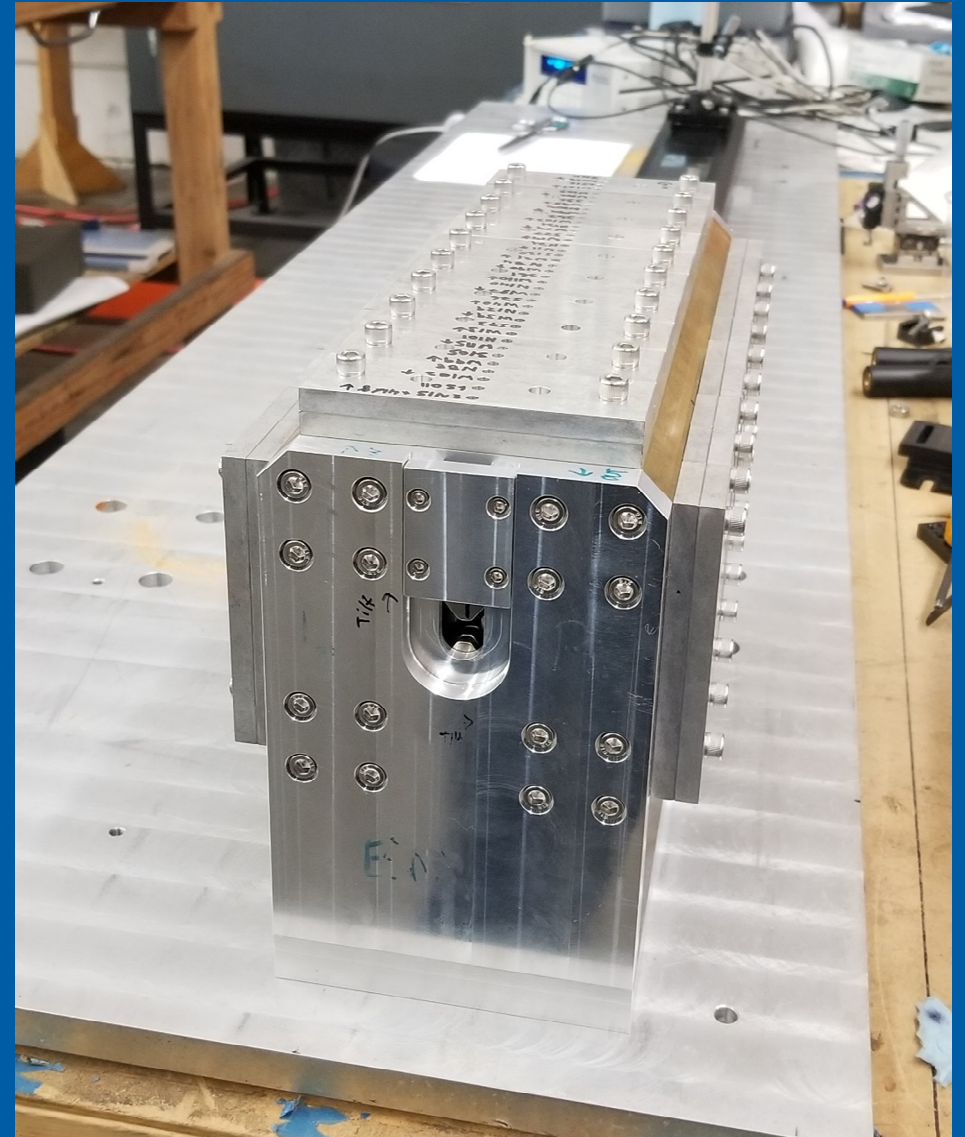
Strongly Tapered Helical Undulator System for TESSA-266

BY TARA CAMPESE
RADIABEAM
SANTA MONICA, CA

The UCLA logo consists of the letters "UCLA" in a bold, blue, sans-serif font, set against a white background.

The Argonne National Laboratory logo features the word "Argonne" in a black serif font, with "NATIONAL LABORATORY" in a smaller black sans-serif font below it. To the right is a colorful geometric logo made of four triangles (green, red, blue, and yellow) forming a larger triangle.

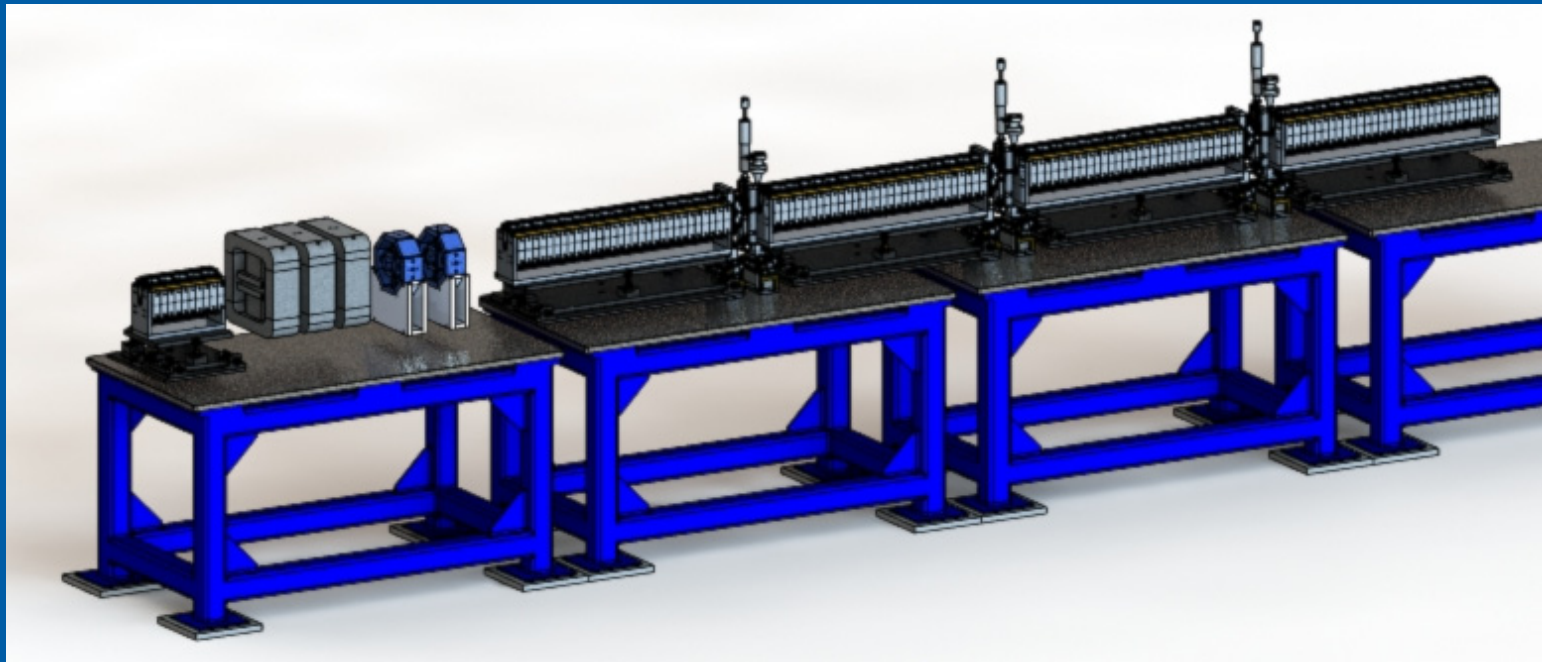
- Introduction
- THESEUS: Tapered HELical Segmented Undulator System
- Magnets and Magnet Holders
- Design Consideration
- Chamber
- Initial Measurement
- Precise Measurements
- Break Section
- Tunable Quadrupole
- Future Work
- Acknowledgements
- References
- Questions



- RadiaBeam – Engineering and Manufacturing
- UCLA – Design
- Argonne National Lab (ANL) – Facility
 - Experiment to be performed at the Advanced Photo Source (APS) Linac Extension Area (LEA)

Goals

- Increase Free-Electron-Laser (FEL) efficiency using prebunched electron beams, intense seed laser, and strongly tapered undulators (TESSA^[1] scheme)



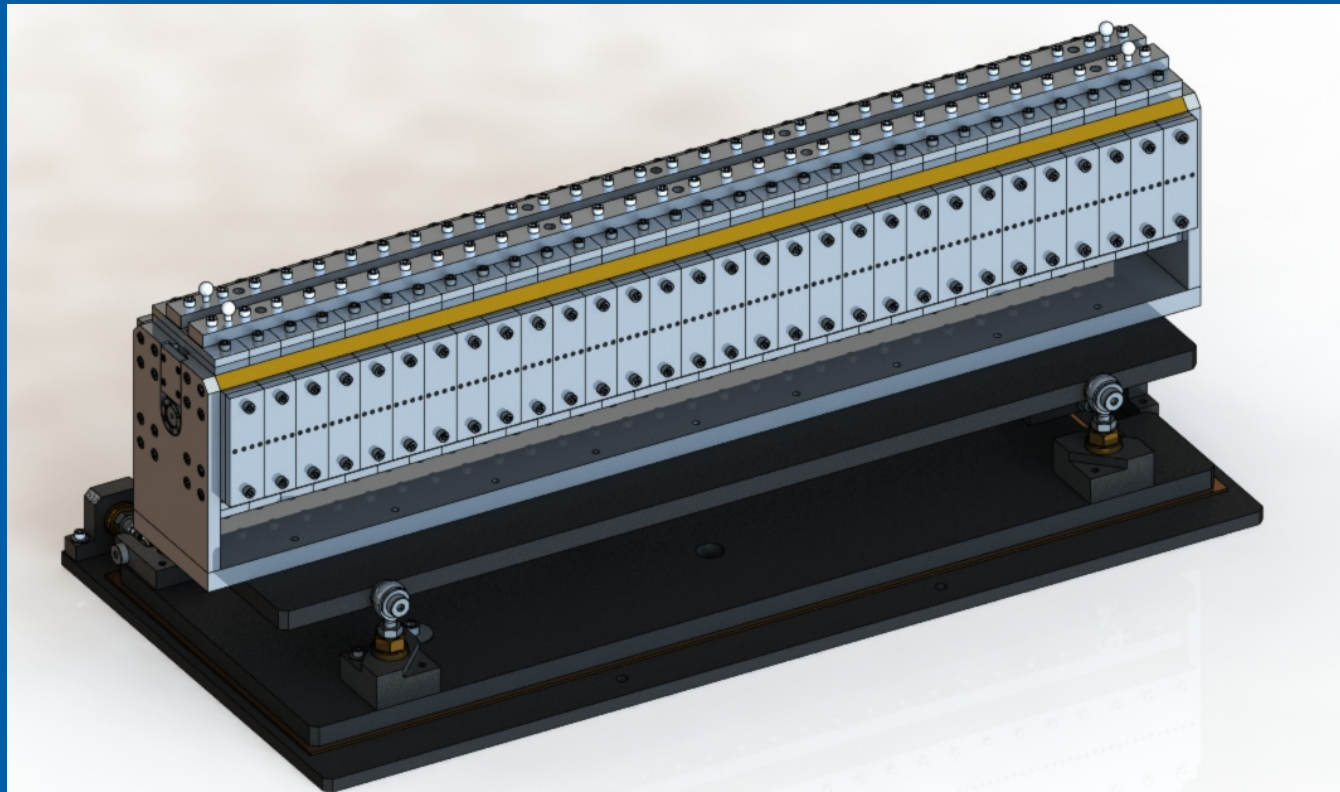
From left to right: Prebuncher, Chicane (3 Dipoles), Quadrupole Doublet, THESEUS 1, Break Section Assembly 1 (Quadrupole, Phase Shifter Dipole, Diagnostic, Quadrupole), THESEUS 2, Break Section 2, THESEUS 3, Break Section 3, and THESEUS 4

TESSA Parameters

Beam Energy	375 MeV
Peak Current	1kA
Emittance	2um
Energy Spread	0.1%
RMS Mean Spot Size	47um
Radiation Wavelength	266nm
Seed Power	1GW

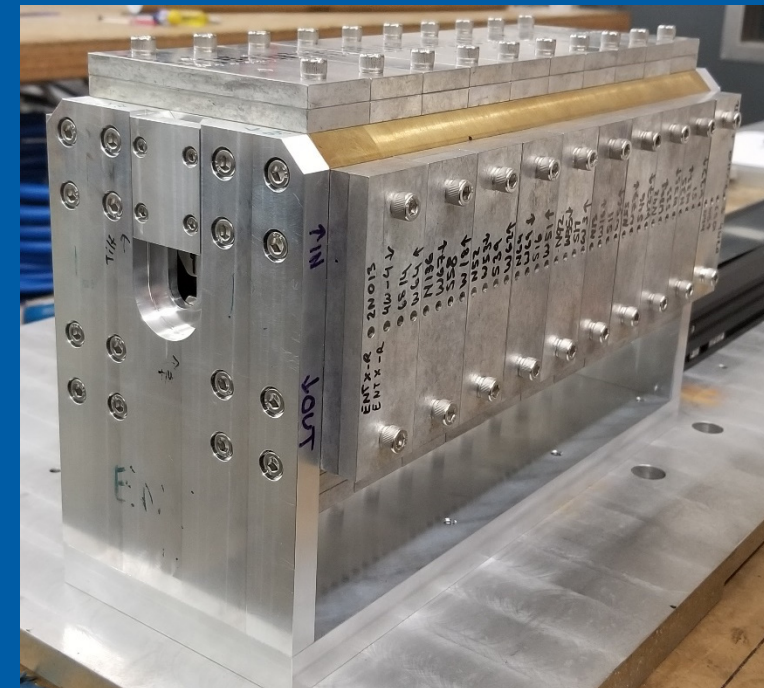
Tapered HELical Segmented Undulator System^[2]

- Undulator length: 989.4 mm
- Prebuncher length: 349.4 mm



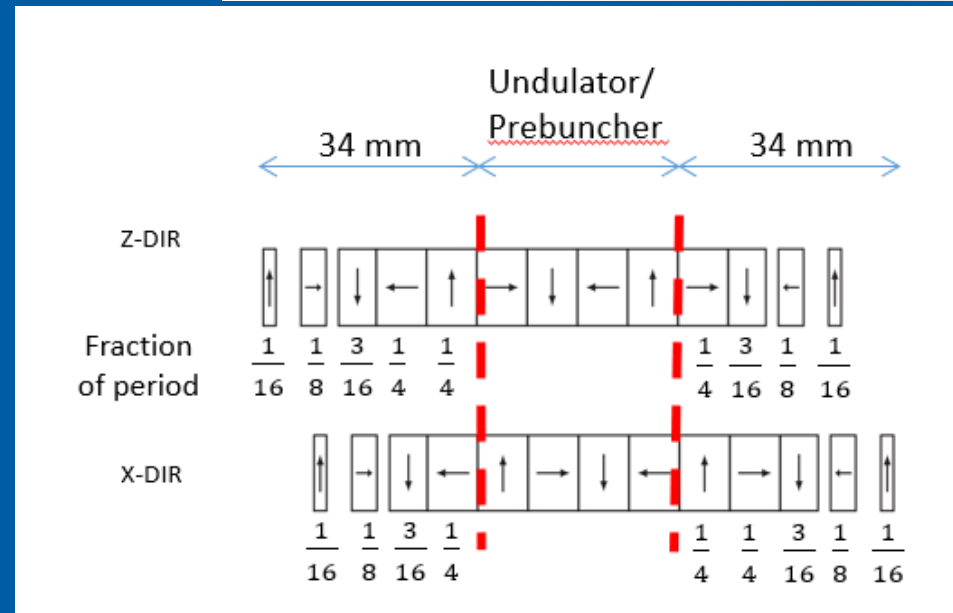
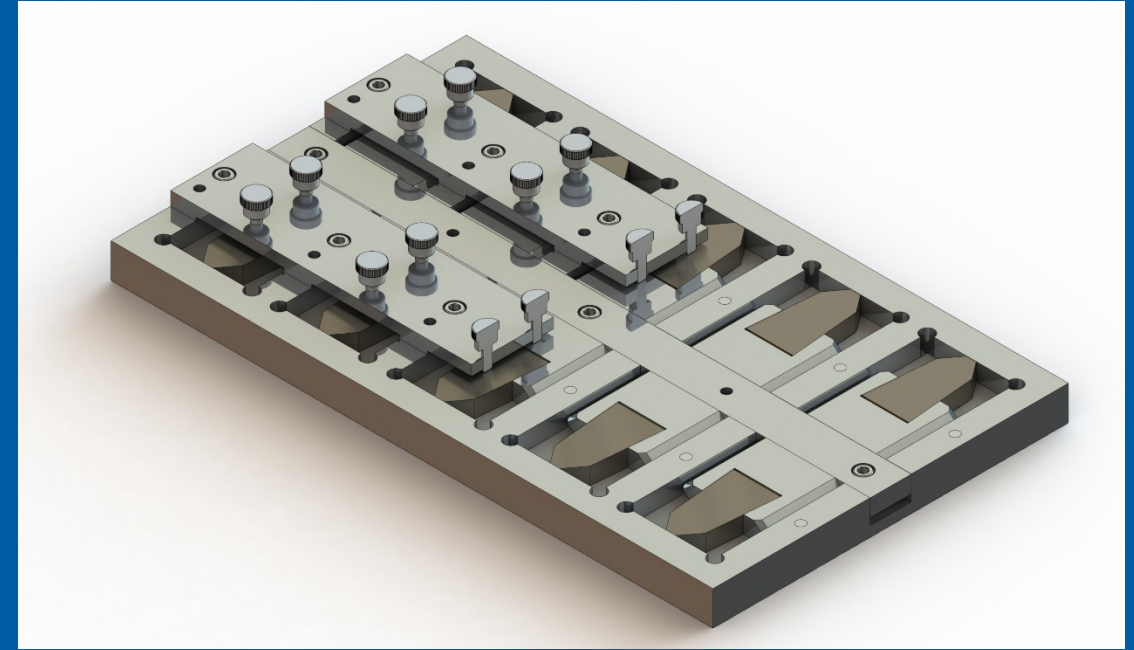
THESEUS Parameters

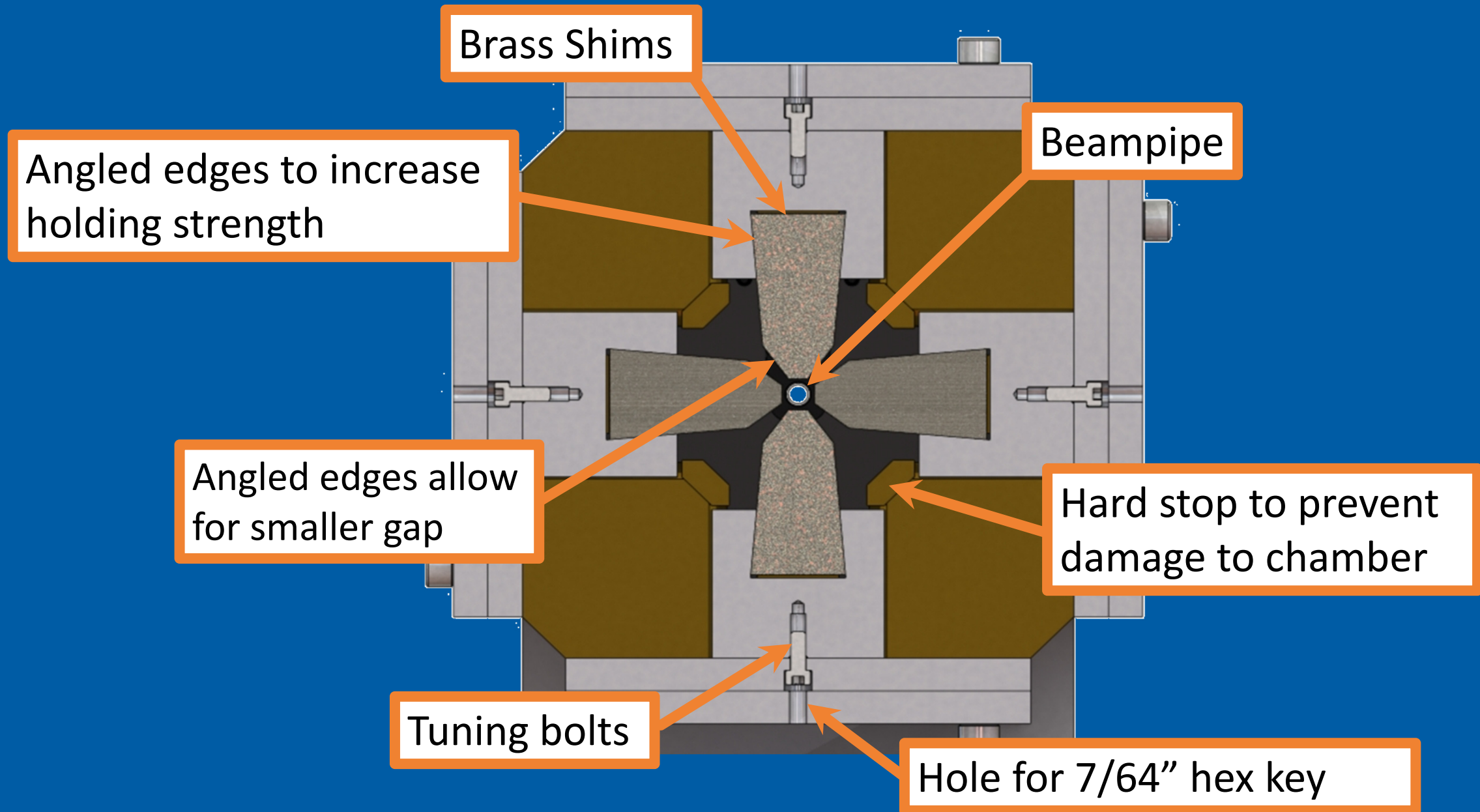
Period Length	32 mm
Magnet Gap (Minimum)	5.58 mm
Number of Full Periods	
Prebuncher	8
Undulator	28
Undulator K Parameter	2.8-2.4



Magnets and Magnet Holders

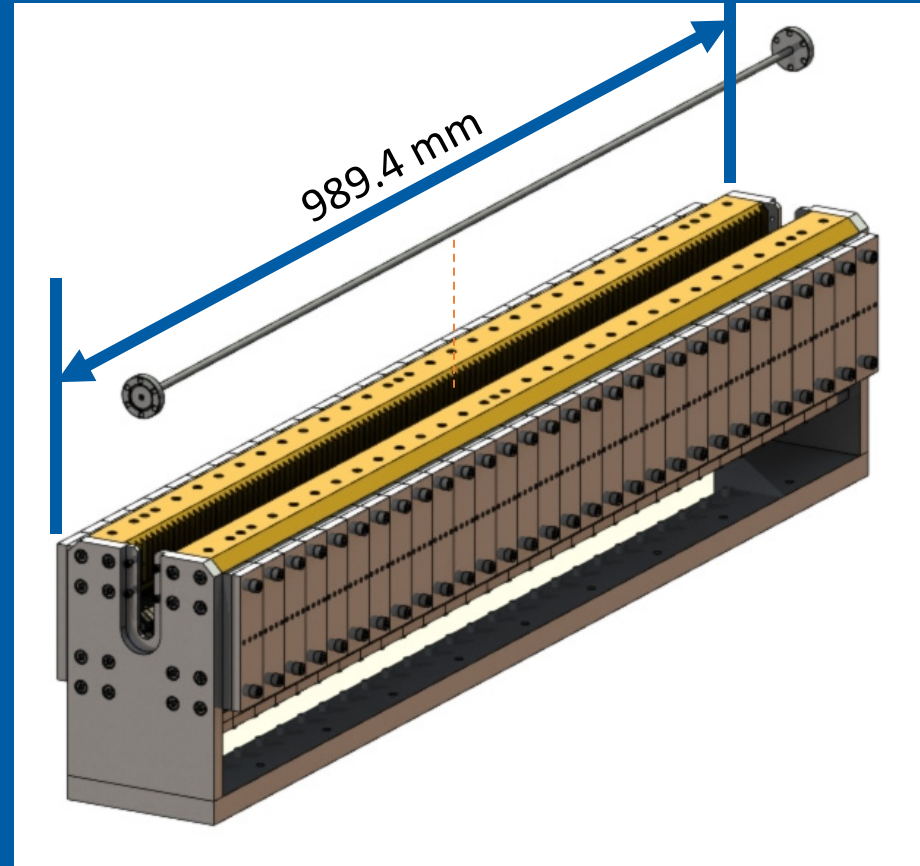
- 8 mm (1/4 period) magnets
- Entrance/Exit Magnets
 - 6 mm (3/16 period)
 - 4 mm (1/8 period)
 - 2 mm (1/16 period)
- Gluing
 - 2-Part Epoxy: Loctite M-31CL^[4]
 - Shim magnets in fixture
- Individually Tunable*
 - 100 μm of travel per 1/8 revolution of tuning bolts



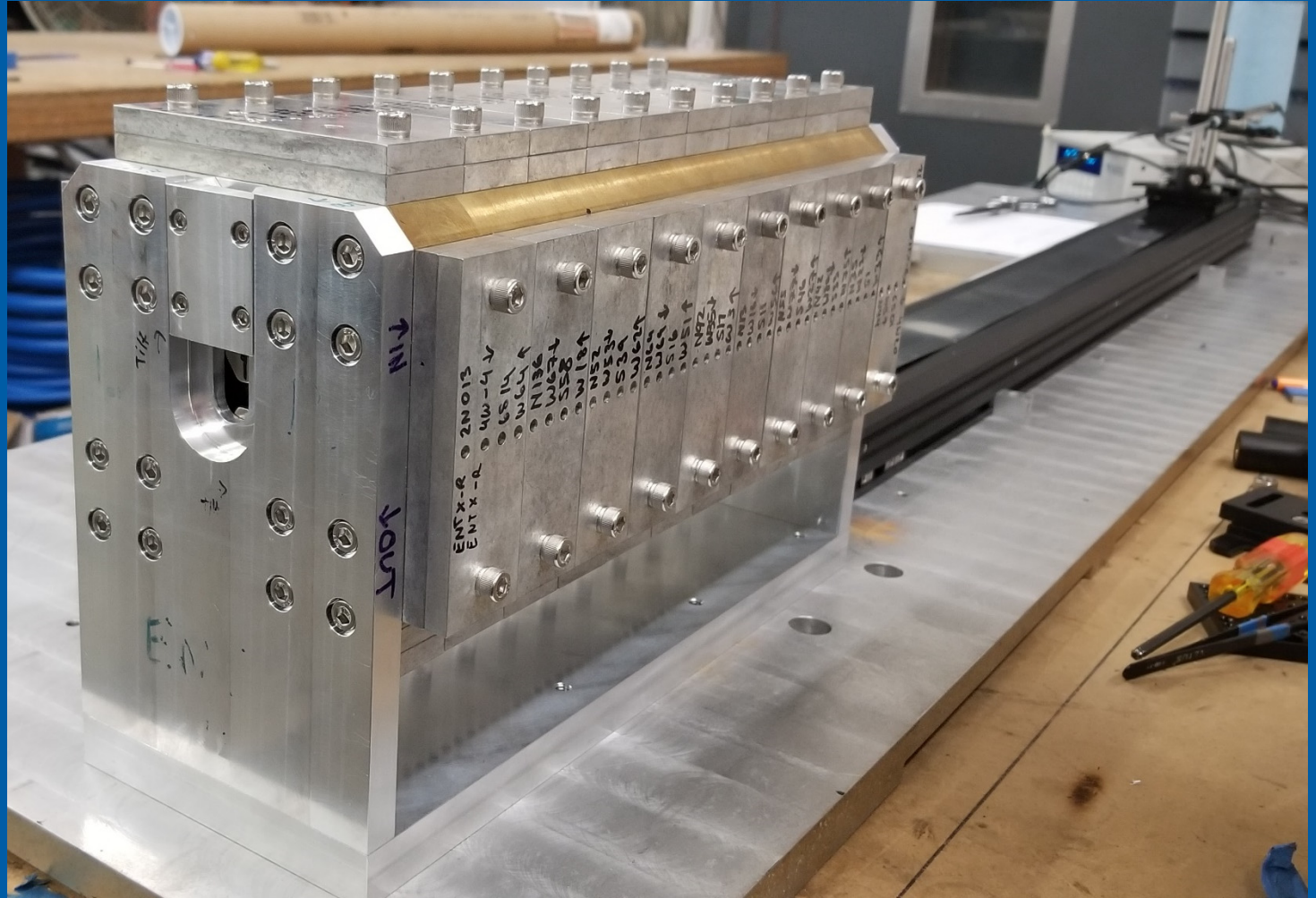
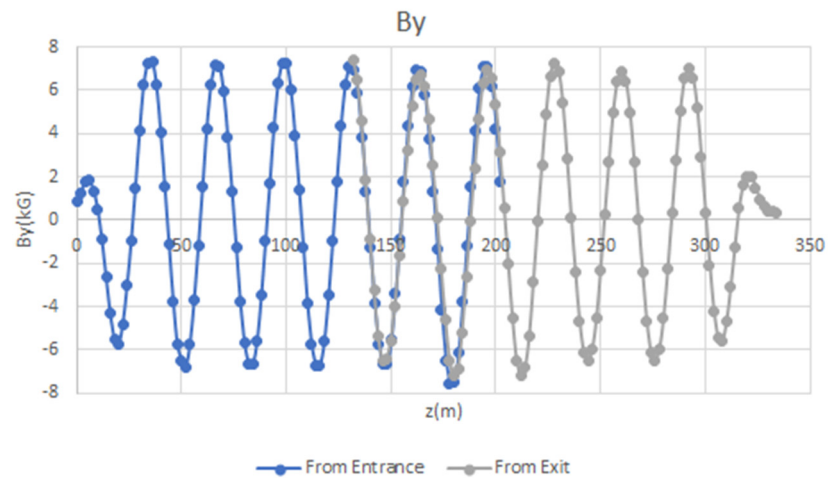
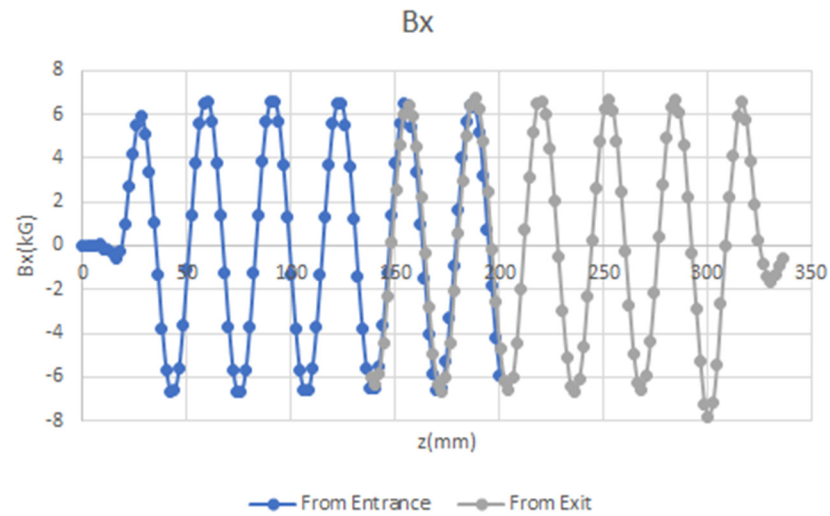


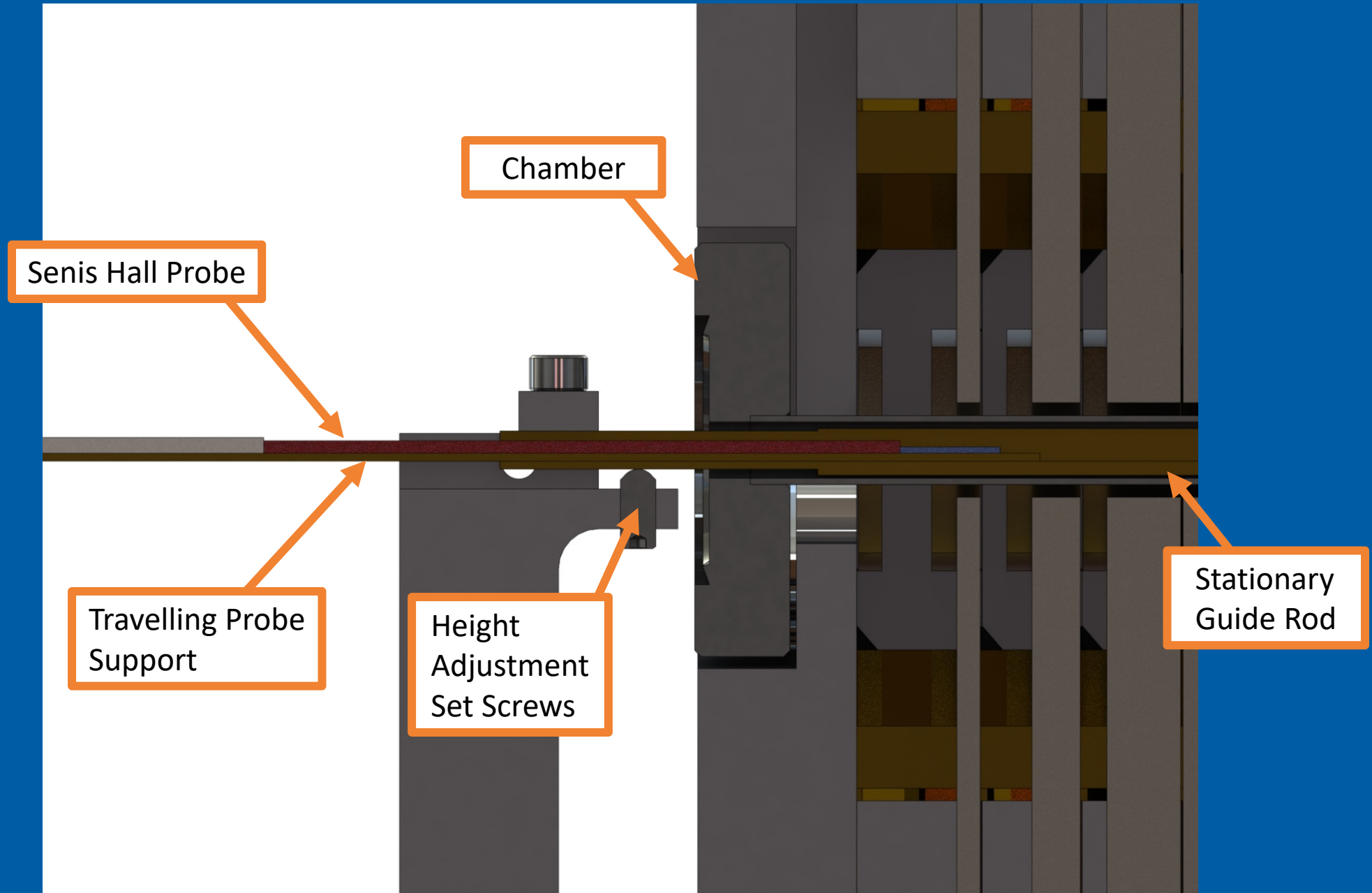
- Chamber

- 316 L stainless steel
- 7/32" [5.55mm] OD x .020" [0.5mm] wall thickness
- 63-125 ra internal surface roughness
- 2x fixed, tapped, 316 LN stainless steel, 1.33" CF flanges
- Inserted from top
- Magnetic permeability limit of 1.05 with exception of weld joints



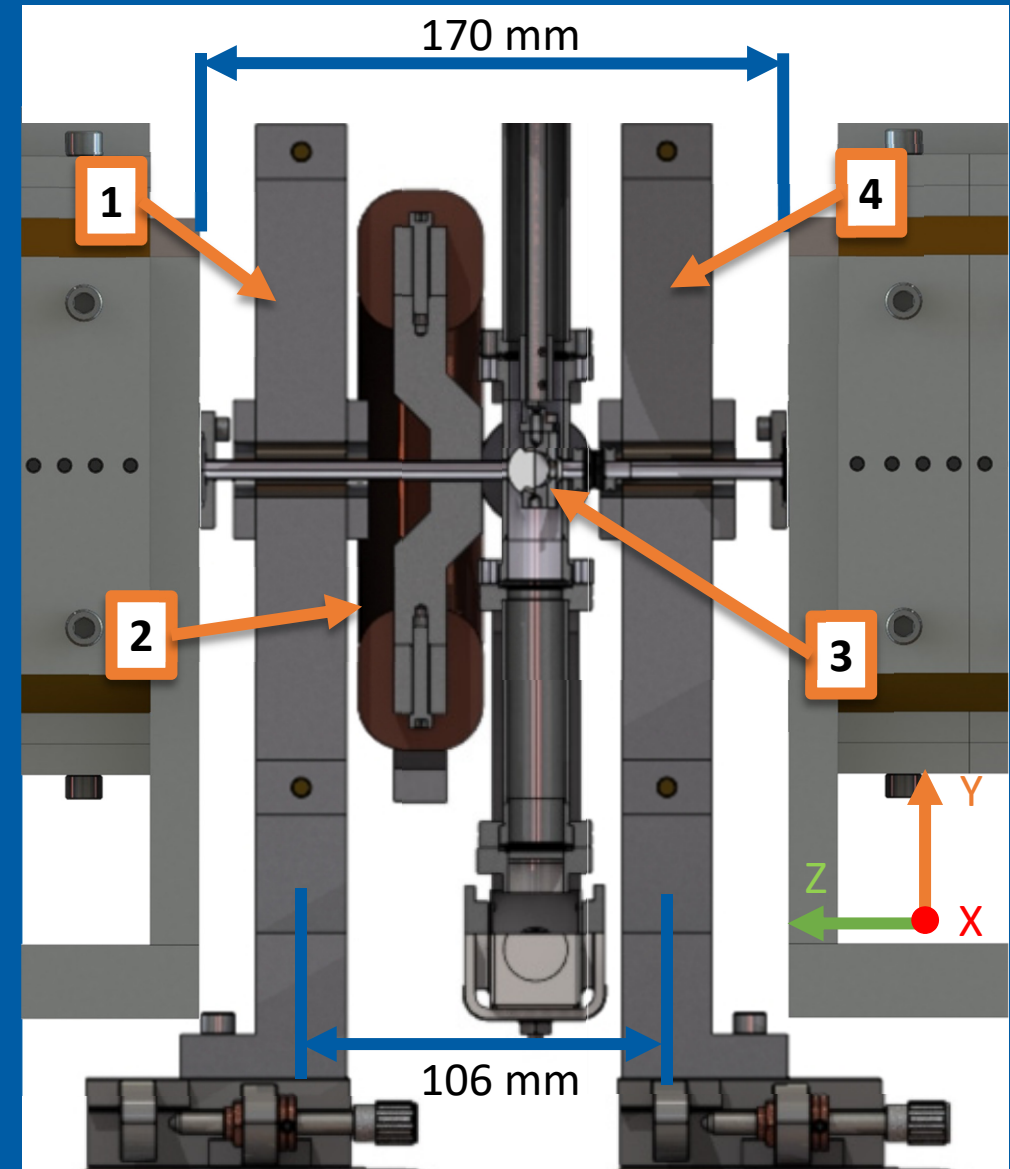
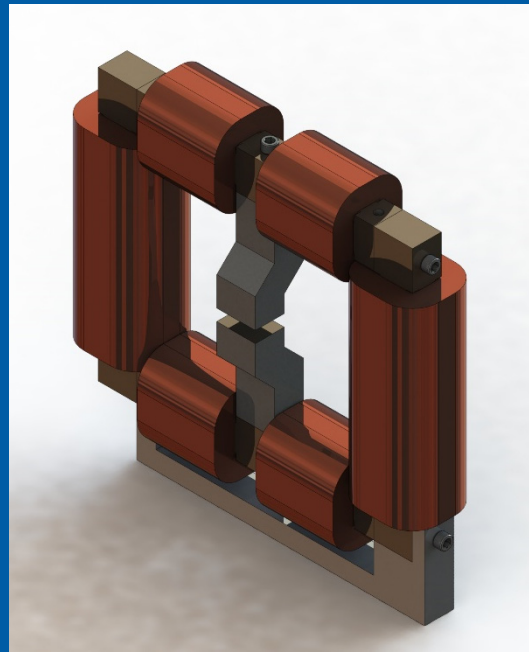
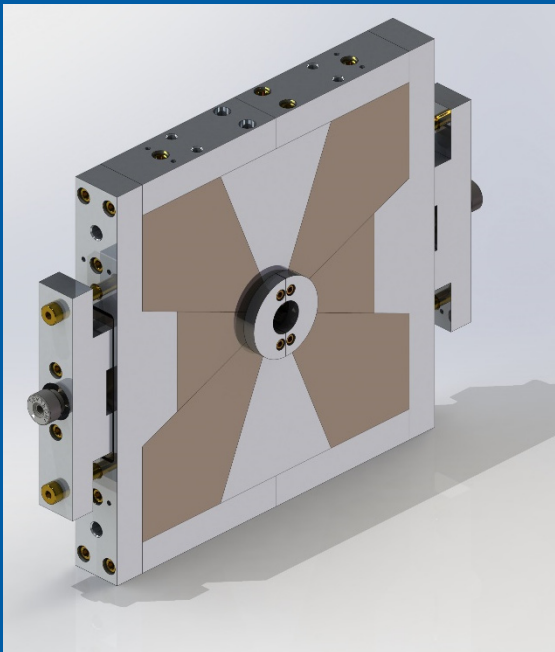
Initial Measurement





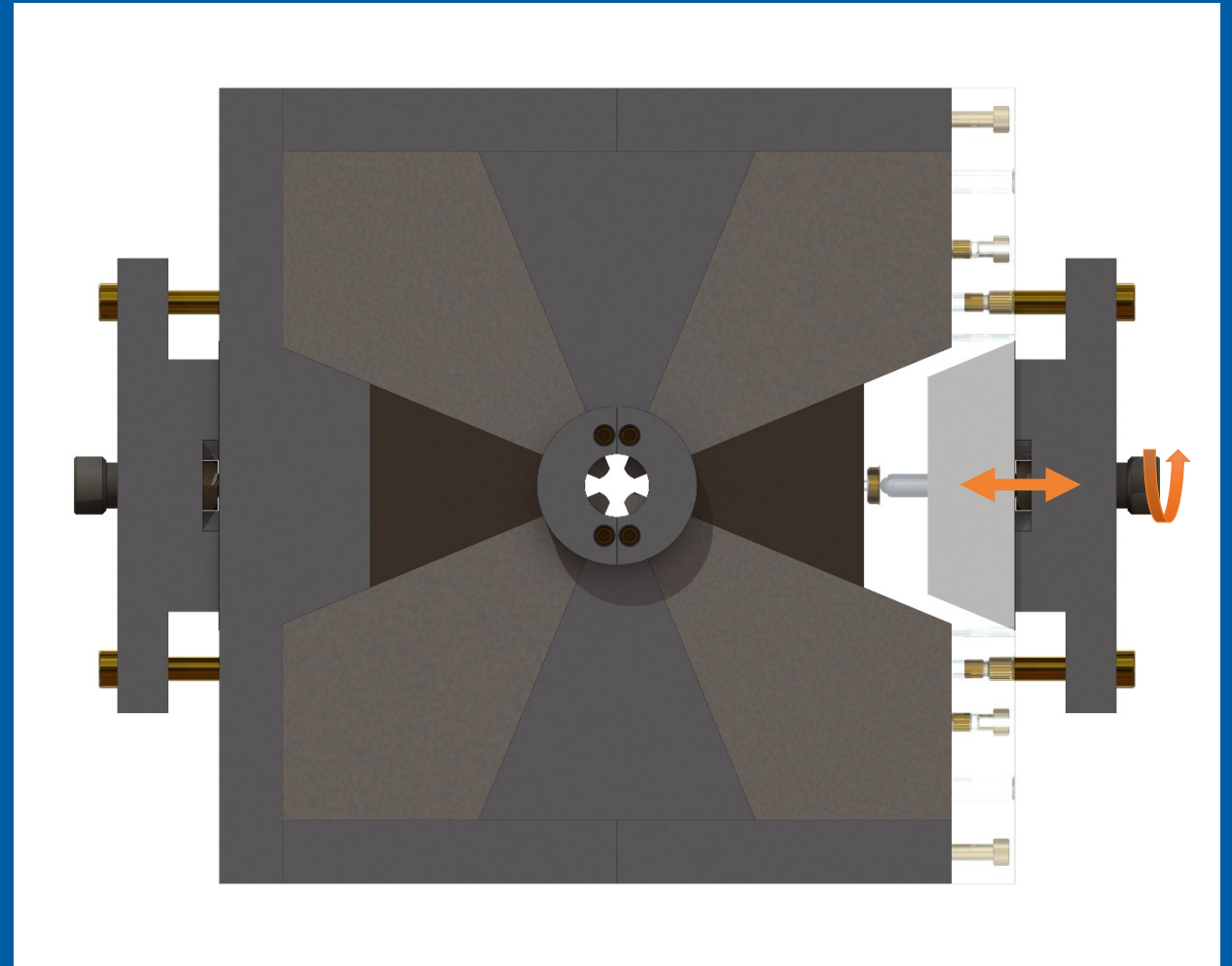
▪ Break Section Components:

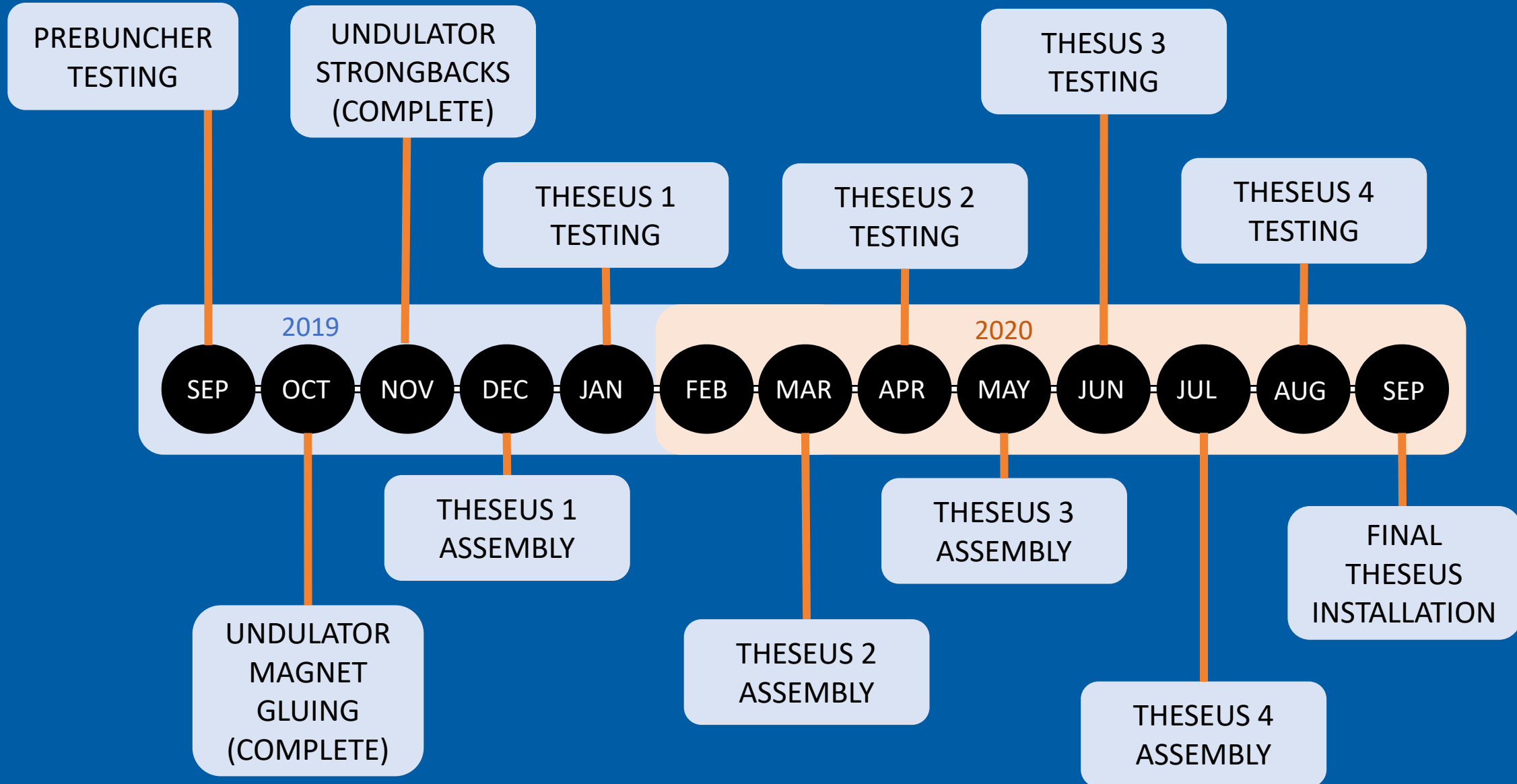
1. 26.2 mm Permanent Magnet Quadrupole (Tunable)
2. 15 mm Electromagnetic Dipole
3. YAG/OTR Combination
4. 26.2 mm Permanent Magnet Quadrupole (Tunable)



Tunable Quadrupole

- Requirements
 - Shim adjustable from 5 mm-15 mm
 - Shim adjustability resolution: ≤ 0.1 mm
 - Install around beampipe
 - Physical length: 26.2 mm
 - Adjustable by stages in X & Y
- Pole tip geometry defined by $y = \frac{r_i^2}{2x}$
- ID: 8mm
- Magnets: NdFeB Grade N52





Acknowledgements



RadiaBeam

R. Agustsson

I. Gadjev

A. Murokh



University of California, Los Angeles

P. Denham

P. Musumeci

Y. Park



Argonne National Laboratory

W. Berg

A. Zholents

Work supported by DOE SBIR Award No. DE-SC0017102

- [1] J. Duris, A. Murokh, and P. Musumeci. "Tapering enhanced stimulated superradiant amplification." *New Journal of Physics*, vol. 17, no. 6, p. 063036, Jun. 2015.
- [2] Y. Park *et al.*, "High Efficiency and High Gain Amplification at 266 nm Wavelength", presented at the 39th Int. Free Electron Laser Conf. (FEL'19), Hamburg, Germany, Aug. 2019, paper THP073.
- [3] J. Duris *et al.* "High-quality electron beams from a helical inverse free-electron laser accelerator." *Nature communications*, vol. 5, article 4928, Sep. 2014.
- [4] LOCTITE Hysol M031CL Datasheet, <https://static.rshughes.com/wm/p/wm-asis/d4c9ef924f355becafa54d8600e121a0586b31fc.pdf>

